

# EXPERIMENT - 2

## Title

Study and Implementation of SQL Fundamentals, DDL, DML Commands, Constraints, Data Types, SELECT Queries, Filtering, Sorting and Aggregate Functions

## Aim

To learn and implement SQL fundamentals including DDL and DML commands, constraints, data types, SELECT queries with filtering and sorting, and aggregate functions in a database.

## Relational Schema

### Student

- StudentID (PK)
- StudentName
- Email (UNIQUE)
- DeptID (FK)

### Department

- DeptID (PK)
- DeptName

### Course

- CourseID (PK)
- CourseName
- Credits

### Enrollment

- EnrollID (PK)
- StudentID (FK)
- CourseID (FK)

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## Source Code (MySQL)

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Department (
```

```
    DeptID INT PRIMARY KEY,
```

```
    DeptName VARCHAR(50) NOT NULL UNIQUE
```

```
);
```

```
CREATE TABLE Student (
```

```
    StudentID INT PRIMARY KEY,
```

```
    StudentName VARCHAR(50) NOT NULL,
```

```
    Email VARCHAR(100) UNIQUE,
```

```
    DeptID INT,
```

```
    FOREIGN KEY (DeptID)
```

```
    REFERENCES Department(DeptID)
```

```
    ON DELETE SET NULL
```

```
);
```

```
CREATE TABLE Course (
```

```
    CourseID INT PRIMARY KEY,
```

```
    CourseName VARCHAR(50) NOT NULL,
```

```
    Credits INT NOT NULL
```

```
);
```

```
CREATE TABLE Enrollment (
```

```
    EnrollID INT PRIMARY KEY,
```

```
    StudentID INT,
```

```
    CourseID INT,
```

```
    FOREIGN KEY (StudentID)
```

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REFERENCES Student(StudentID)

ON DELETE CASCADE,

FOREIGN KEY (CourseID)

REFERENCES Course(CourseID)

ON DELETE CASCADE

);

### Output

#### Department

| DeptID | DeptName |
|--------|----------|
| 1      | CSE      |
| 2      | AIML     |

#### Student

| StudentID | StudentName | Email           | DeptID |
|-----------|-------------|-----------------|--------|
| 101       | Rahul       | rahul@gmail.com | 1      |
| 102       | Priya       | priya@gmail.com | 2      |

#### Course

| CourseID | CourseName       | Credits |
|----------|------------------|---------|
| 201      | DBMS             | 4       |
| 202      | Operating System | 4       |

#### Enrollment

| EnrollID | StudentID | CourseID |
|----------|-----------|----------|
| 1        | 101       | 201      |
| 2        | 102       | 202      |